

Project Outline:

Sentencing Disparities Between Sexes

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16 June 2025

1 Introduction and Motivation

Investigating claims on disparities between sexes for sentencing is important because it highlights potential flaws in our justice system. The justice system should operate under the notion that everyone is innocent until proven guilty. All individuals deserve a fair trial and equitable treatment, as public trust depends on the court's commitment to due diligence and justice for all.

In this project, my goal is to investigate the claim that women are sentenced more harshly than men for homicide cases in Florida and to create a scalable, automated framework that can apply parameters to other types of offenses.

2 Literature Review and Institutional Details

Here is a list of academic papers that have explored biases that occur in our judicial system (primarily based on race) or have explained the theories behind economical and statistical discrimination.

2.1 A Test of Racial Bias in Capital Sentencing

Authors: Alberto Alesina and Eliana La Ferrara

Publication: American Economic Review, 2014

Objective: To investigate whether racial bias affects capital sentencing decisions in the United States by analyzing patterns of sentence reversals on appeal.

Summary: The paper models courts as decision-makers minimizing errors (wrongful convictions or wrongful acquittals) and examines whether error rates differ by race of defendant and victim. The authors find evidence of racial bias particularly in Southern states, with minority defendants who kill white victims facing higher reversal rates, suggesting unequal treatment in the justice system.

Relevance to Project: This study's approach to modeling judicial decisions and testing for bias using error patterns informs the theoretical framework and empirical strategy for detecting sex-based sentencing

disparities in homicide cases. Its focus on detailed sentencing outcomes and subgroup analysis also highlights the importance of controlling for context to avoid misleading aggregate conclusions.

2.2 A Review of Thomas Sowell's *Discrimination and Disparities*

Authors: Jennifer L. Doleac

Publication: The Journal of Economic Literature, 2021

Objective: To critically assess Thomas Sowell's arguments about the causes of disparities across groups, including discrimination, culture, and other factors.

Summary: Doleac summarizes Sowell's view that disparities arise from multiple sources beyond discrimination, such as cultural differences and choices. She discusses how Sowell challenges common assumptions about systemic bias and emphasizes the complexity of interpreting disparities. The review highlights the nuanced debate on causes of inequality.

Relevance to Project: This review provides important context for understanding sentencing disparities by emphasizing that differences may arise from various factors, not solely bias—helping frame the analysis with caution against oversimplified conclusions.

2.3 The Economics of Discrimination

Authors: Gary S. Becker

Publication: The University of Chicago Press, 1957

Objective: To model how individual preferences or “tastes” against certain groups lead to discriminatory behavior in markets, despite economic incentives.

Summary: Becker introduces the concept that some employers, customers, or employees have a “disutility” or distaste for interacting with certain groups (based on race, sex, etc.). This prejudice causes them to pay a cost to avoid hiring or dealing with these groups, leading to inefficient market outcomes and wage or sentencing disparities.

Relevance to Project: Becker's model helps interpret sentencing disparities as possibly stemming from judges' personal biases or stereotypes (taste-based discrimination), rather than purely statistical or rational decision-making.

2.4 The Theory of Discrimination

Authors: Kenneth Arrow

Publication: Industrial Relations Section of Princeton University, 1971

Objective: To explain how discrimination can persist in markets due to imperfect information, where decision-makers use observable group characteristics as proxies for unobserved traits.

Summary: Arrow develops a model of statistical discrimination showing that even rational agents without personal prejudice may discriminate when they rely on group averages to predict individual outcomes. This leads to persistent disparities, as decisions are based on imperfect signals rather than individual merit.

Relevance to Project: This theory provides a foundation for interpreting sentencing disparities as potentially arising from judges using sex as a proxy for unobserved risk factors, helping to distinguish between taste-based and statistical discrimination in criminal justice outcomes.

2.5 The Economics of Discrimination: Economists Enter the Courtroom

Authors: Orley Ashenfelter and Ronald Oaxaca.

Publication: The American Economic Review, 1987.

Objective: To show how economists use tools like the Oaxaca–Blinder decomposition to analyze race and gender discrimination in legal civil cases.

Summary: The paper explains how outcome disparities (in wages) can be broken into explained vs. unexplained components, with the latter potentially signaling discrimination. It highlights how courts have used this analysis in real cases.

Relevance to Project: I can apply the same decomposition method to sentencing outcomes in homicide cases to detect sex-based disparities, extending its approach from civil to criminal law.

2.6 Examiner and Judge Designs in Economics: A Practitioner’s Guide

Authors: Eric Chyn, Brigham Frandsen, and Emily Leslie

Publication: The Journal of Economic Literature, 2025

Objective: To provide a practical overview of “examiner” and “judge” research designs that exploit variation in decision-makers’ tendencies to identify causal effects in economics, with applications to legal settings.

Summary: The authors review methodologies that leverage differences in individual adjudicators’ treatment propensities as sources of quasi-random variation. These designs help isolate the causal impact of judicial decisions by controlling for case characteristics and exploiting the randomness in judge assignment. An empirical application to pre-trial detention demonstrates the approach’s utility for identifying effects of judicial discretion.

Relevance to Project: This paper’s framework for exploiting judge-level heterogeneity can complement traditional sentencing analyses by accounting for decision-maker variation, potentially improving causal inference on sex-based sentencing disparities. Incorporating such designs could help address concerns about unobserved judicial heterogeneity affecting sentencing outcomes in homicide cases.

3 Theoretical Model

3.1 Statistical Decision Theory and Sentencing as Estimation

Judicial sentencing decisions can be understood through the lens of statistical decision theory. A judge is modeled as a decision-maker operating under uncertainty, seeking to minimize a loss function that weighs the costs of judicial error. In this context, two types of error are possible: Type I errors represent overly harsh sentences, while Type II errors reflect excessively lenient punishments that may compromise public safety.

Building on the framework of Alesina and La Ferrara (2014), sentencing is treated as a probabilistic estimate of a defendant’s latent culpability or social risk, conditional on observed legal and demographic variables. Sentencing outcomes are constrained by statutory minimum and maximum bounds. To standardize severity across different offenses, the actual sentence is modeled as a scaled function within these legal limits:

$$\text{Sentence}_i = \text{Min}_i + (\text{Max}_i - \text{Min}_i) \cdot b_i$$

Here, $b_i \in [0, 1]$ represents the judge’s implicit belief about how severe the sentence should be, relative to the legal range. The normalized sentencing outcome is then log-transformed as:

$$y_i = \log \left(\frac{\text{Sentence}_i - \text{Min}_i}{\text{Max}_i - \text{Min}_i} \right)$$

This transformation produces a continuous, unbounded measure of relative sentence severity, allowing for meaningful comparisons across different charge types. Life sentences, which do not vary in length and lie outside this bounded framework, are handled separately as a binary outcome.

3.2 Economic Discrimination Theory

This study is also guided by two major theoretical models from the economics of discrimination. The first, developed by Becker (1957), posits that some decision-makers exhibit "taste-based" discrimination—that is, they act on personal biases or stereotypes that lead to differential treatment, even when such differences are legally irrelevant.

The second, presented by Arrow (1971), focuses on statistical discrimination. In this framework, judges use observable characteristics (such as sex) as proxies for unobserved traits like remorse, risk of recidivism, or danger to society. While not necessarily motivated by prejudice, such behavior can still produce systematic disparities in outcomes. These models help frame potential explanations for sex-based differences in sentencing decisions.

3.3 Oaxaca–Blinder Decomposition with Log Outcomes

To empirically distinguish between disparities arising from observable case differences and those due to unequal treatment, I will apply the Oaxaca–Blinder decomposition. This technique decomposes the mean gap in sentencing outcomes between men and women into an "explained" component, reflecting differences in characteristics, and an "unexplained" component, which may reflect discrimination or unobserved factors.

Separate regressions are estimated for male and female defendants:

$$\log y_M = X_M \hat{\beta}_M, \quad \log y_F = X_F \hat{\beta}_F$$

The difference in average log sentencing is decomposed as:

$$\log \bar{y}_M - \log \bar{y}_F = (\bar{X}_M - \bar{X}_F)\hat{\beta}_M + \bar{X}_F(\hat{\beta}_M - \hat{\beta}_F)$$

The first term represents the explained difference due to characteristics like charge type, prior records, use of a public defender and so on. The second term captures the unexplained gap, which could signal differential treatment across sexes.

4 Empirical Specification

4.1 Sentencing Outcome Definition

The sentencing process is modeled as a two-stage framework. First, the judge decides whether to impose a life sentence. If not, a sentence is chosen from the allowable statutory range. These two outcomes are estimated separately to accommodate their distinct statistical structures.

For defendants not sentenced to life, the primary outcome is a normalized and log-transformed sentencing ratio:

$$y_i = \log \left(\frac{\text{Sentence}_i - \text{Min}_i}{\text{Max}_i - \text{Min}_i} \right)$$

This transformation captures proportional differences in sentencing severity while controlling for statutory bounds. Cases involving life sentences are excluded from this specification due to undefined minimum and maximum values.

Life sentences are modeled as a binary outcome:

$$\text{Life}_{i,c} = \begin{cases} 1 & \text{if life sentence imposed} \\ 0 & \text{otherwise} \end{cases}$$

This distinction allows the analysis to account for the qualitative difference between determinate sentencing and life imprisonment.

4.2 Estimation Strategy

Life Sentencing is modeled using a probit specification:

$$\Pr(\text{Life}_i = 1 \mid X_i) = \Phi \left(\frac{X_i \beta}{\sigma} \right)$$

where X_i includes sex and other relevant controls, and Φ is the standard normal cumulative distribution function.

Non-Life Sentencing among those who do not receive life sentences, sentencing severity is estimated with a linear model:

$$y_i = \beta_0 + \beta_1 S_i + \beta_2^\top X_i + \varepsilon_i$$

Here, S_i indicates sex, X_i is a set of covariates, and ε_i is the residual.

If exploratory analysis reveals that the outcome distribution deviates significantly from normality or shows signs of heteroskedasticity, a **Generalized Linear Model** may be adopted. In particular, a Beta regression with a logit link could be appropriate if the normalized sentencing ratio is found to follow a Beta-like distribution:

$$E[b_i | X_i] = \mu_i, \quad \text{logit}(\mu_i) = X_i \beta$$

This offers a flexible framework for modeling sentencing within bounded intervals, accommodating asymmetry and skewness.

Selection Consideration: One empirical concern is that disparities in sentence length may be partly driven by selection into life sentencing. For example, if more men receive life sentences, average non-life sentences among women could appear higher due to this selection process. This motivates the two-part modeling framework and may be further investigated through sample selection or subgroup analysis.

4.3 Covariates

All models will control for a rich set of covariates, including defendant demographics, charge degree, prior criminal history, victim information (if available), county, plea status, and type of legal representation. These controls ensure that observed disparities are not simply driven by compositional differences in case severity or context.

4.4 Decomposition and Subgroup Analysis

To further unpack the sources of any observed disparities, Oaxaca–Blinder decompositions will be performed across subgroups. For example, decomposition may be run separately for defendants who used public versus private defense attorneys, or across different degrees of homicide charges. This allows for identification of whether sex-based disparities are concentrated in specific contexts, and mitigates concerns about aggregation

bias or Simpson’s paradox.

4.5 Scalability

While this study focuses on homicide cases in Florida, the framework is designed to be extensible. Once validated, the methods can be adapted to other charge categories by updating offense-specific parameters. The ultimate aim is to develop a scalable system for automated detection of sentencing disparities throughout the criminal code.

5 Data

5.1 Data Sets and Characteristics

The data used in this study comprises detailed records on hundreds of thousands of cases from Florida’s judicial system. It includes information on arrest events, defendant profiles (such as sex offender status, gang affiliation, and repeat offenses), county locations, and charges filed. The data originates from three main sources, each covering different stages and aspects of the criminal justice process.

First, the Arrest and Booking Reports from the County Detention Offices focus primarily on arrest details. These reports contain variables including the arrest and booking dates, types of charges and statutes involved, the county of occurrence, and descriptions of the relevant statutes. Second, the Case Reports from State Attorneys provide data related to the trial proceedings. Key variables include case creation dates, final decision dates, detailed charge information, final outcomes, and victim characteristics. Third, the Case Reports from Clerks of Court capture non-judicial elements such as disposition and sentencing dates, the type of legal counsel (public defender or private attorney), imposed fines, court costs and restitution, as well as sentencing details including confinement terms and statutory maximum and minimum sentences.

5.2 Data Preparation

A significant challenge in working with this judicial data is the lack of a unified, standardized system for case identification across departments. Each source employs unique identifiers specific to its internal records, which do not align across datasets. For example, identifiers assigned by the Clerk of Courts do not correspond to those used by the State Attorney’s office. Consequently, merging these datasets requires an alternative approach beyond exact key matching.

To address this, I employ fuzzy joining techniques. This method merges records based on approximate matches across shared characteristics, rather than exact identifiers. The merging strategy leverages consistent

case attributes such as statute, defendant demographics (including race and sex), charge degree, and location. Additionally, temporal ordering of events serves as an important constraint; for instance, arrest dates must precede booking dates, which in turn must occur before disposition and sentencing dates. To ensure logical consistency, I impose time windows—for example, booking and arrest dates should be within three days of each other. This approach allows for effective linking of disparate data sources despite the absence of common unique identifiers.

5.3 Data Linking Challenges and Evaluation Strategy

An important issue encountered during the fuzzy joining process is a significant reduction in usable data points—up to half of the original records can be lost. This loss occurs because many cases cannot be confidently matched across datasets due to inconsistencies or missing information.

To address this trade-off between data completeness and variable richness, I plan to conduct comparative analyses of two datasets: (1) the merged dataset obtained after fuzzy joining, which contains fewer observations but a richer set of variables across the criminal justice process; and (2) the standalone Clerk of Courts dataset, which retains a larger number of cases but includes fewer variables, primarily focused on sentencing information.

By examining the distributions of key variables such as charge type and sentence severity—both overall and stratified by defendant sex—in each dataset, I will assess the representativeness and potential biases introduced by data loss in the merged sample. This evaluation will inform whether it is preferable to use the more comprehensive but smaller merged dataset or the larger but less detailed Clerk of Courts dataset for subsequent empirical analysis.

5.4 Data Sources

The datasets employed in this study are obtained from the Criminal Justice Data Transparency (CJDT) framework, maintained by the Florida Department of Law Enforcement. The CJDT provides public access to extensive criminal justice data, supporting transparency and research on the state’s judicial system (Florida Department of Law Enforcement, 2025).

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